

Final Project 2025

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Analyzing Education Career Success data set

For this project we are going to analyze the "education_career_success.csv" data set to see if there are any factors that contribute to obtaining a job offer such as internships, age, gender, university gpa, or certifications to name a few. We specifically want to find out if having one or more internships has an impact on obtaining a job offer.

To do this we state that our null hypothesis, H_0 , infers that one or more completed internships does not have an impact on obtaining a job offer.

The alterernate hypothesis , H_a , states that one or more completed internships does have an impact on obtaining a job offer.

Descriptive Analytics

First, we start by importing the different libraries to read the csv file and perform the descriptive and predictive analysis.

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
import statsmodels.formula.api as sm_api
import seaborn as sns

file = './education_career_success.csv'
df = pd.read_csv(file)

describe = df[['Age', 'University_GPA', 'Internships_Completed']] #Selected three columns
print(describe.describe())
```

	Age	University_GPA	Internships_Completed
count	5000.000000	5000.000000	5000.000000
mean	23.442200	3.020028	1.982200
std	3.473712	0.576047	1.408219
min	18.000000	2.000000	0.000000
25%	20.000000	2.520000	1.000000
50%	23.000000	3.030000	2.000000
75%	26.000000	3.510000	3.000000
max	29.000000	4.000000	4.000000

With the code above we are able to see the descriptions for the columns "Age", "University_GPA", and "Internships_Completed" and the information we want to be placing importance on here is the mean for both the university gpa and the number of internships completed as it gives us a baseline on the average amount for both as these are some of the variables we will be measuring to see if they have any effect in receiving a job offer.

```
In [3]: # Descriptive analytics by 'Field of Study' that measures 'Job Offers'
job_offers_describe = df.groupby('Field_of_Study')['Job_Offers'].describe()
```

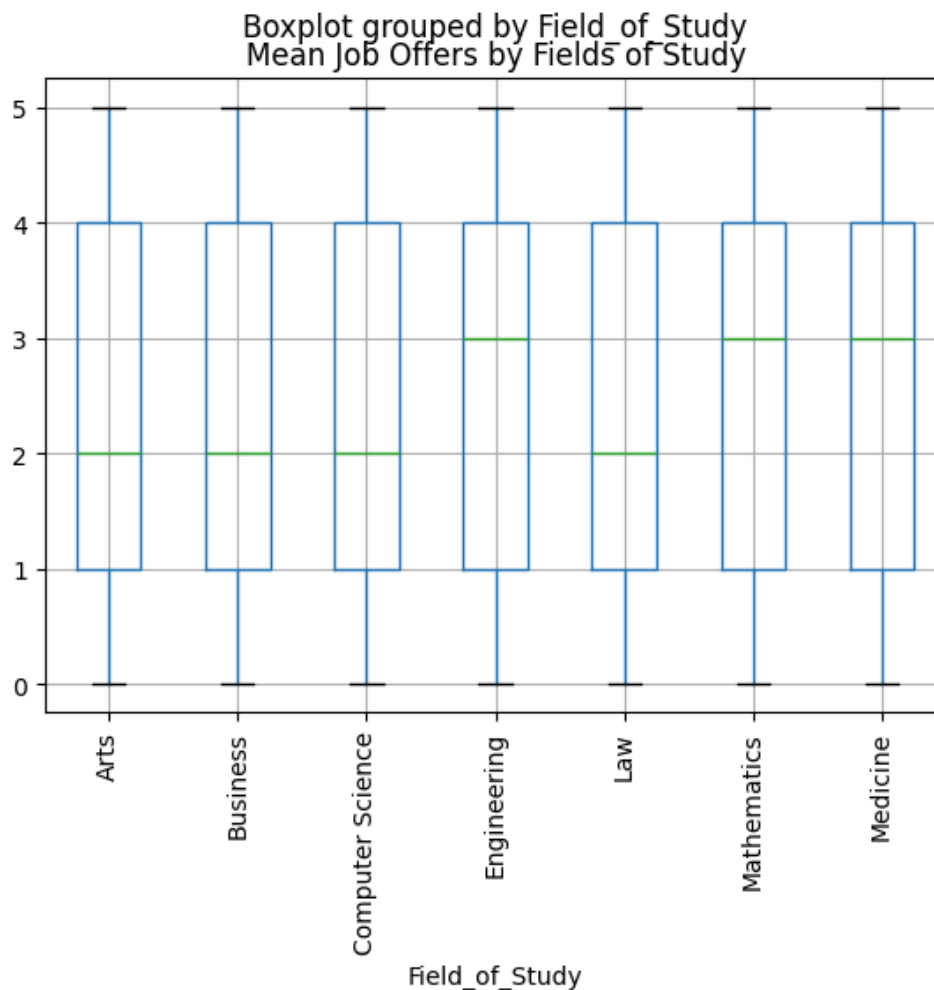
```
print(job_offers_describe)
```

	count	mean	std	min	25%	50%	75%	max
Field_of_Study								
Arts	749.0	2.503338	1.736384	0.0	1.0	2.0	4.0	5.0
Business	719.0	2.453408	1.711907	0.0	1.0	2.0	4.0	5.0
Computer Science	670.0	2.477612	1.695490	0.0	1.0	2.0	4.0	5.0
Engineering	701.0	2.502140	1.693368	0.0	1.0	3.0	4.0	5.0
Law	727.0	2.462173	1.756025	0.0	1.0	2.0	4.0	5.0
Mathematics	745.0	2.514094	1.723533	0.0	1.0	3.0	4.0	5.0
Medicine	689.0	2.507983	1.665399	0.0	1.0	3.0	4.0	5.0

When looking at the fields of study and job offers columns we wanted to ensure there wasn't any major discrepancy within the fields of study that could throw our data off which there wasn't seeing how the mean is within the range of 2.45 and 2.5 remaining relative close to each other.

```
In [4]: # Box plot to visualize the mean Job Offers between different Fields of Study
df.boxplot(column='Job_Offers', by='Field_of_Study')
plt.xticks(rotation = 90)
plt.title('Mean Job Offers by Fields of Study')
```

```
Out[4]: Text(0.5, 1.0, 'Mean Job Offers by Fields of Study')
```



The box plot here helps illustrate our point above helping make clearer how there really isn't much of a difference between fields of study and their propensity to receiving job offers.

```
In [5]: # ANOVA to measure if there is a statistically significant difference in Job Offers between
model = sm.api.ols('Job_Offers ~ C(Field_of_Study)', data=df).fit()
```

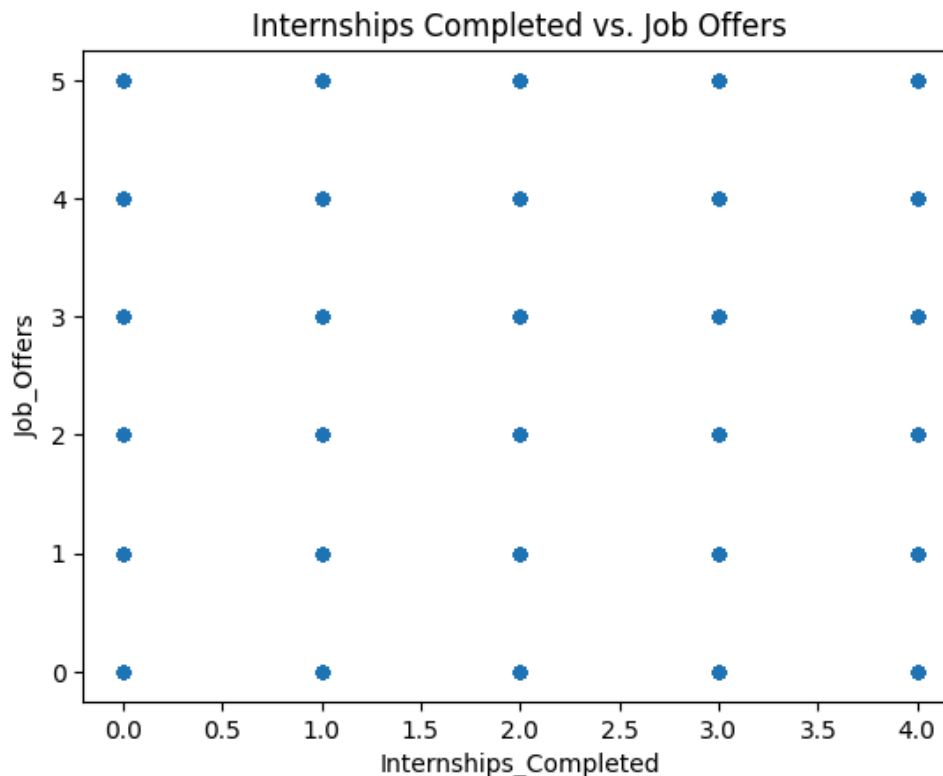
```
anova_table = sm.stats.anova_lm(model, typ=2)
print(anova_table)
```

	sum_sq	df	F	PR(>F)
C(Field_of_Study)	2.513148	6.0	0.142785	0.990458
Residual	14646.859652	4993.0	NaN	NaN

Running the ANOVA for the columns of *Job_Ofers* and *Field_of_Study* and interpreting our results we can see how our p-value is higher than the standard .05 thus we can conclude that there is no statistically significant difference in *Job Offers* between the different *Fields of Study*.

```
In [6]: # Scatter plot to show relationship between Internships and Job Offers
df.plot.scatter(x='Internships_Completed', y='Job_Offers')
plt.title('Internships Completed vs. Job Offers')
```

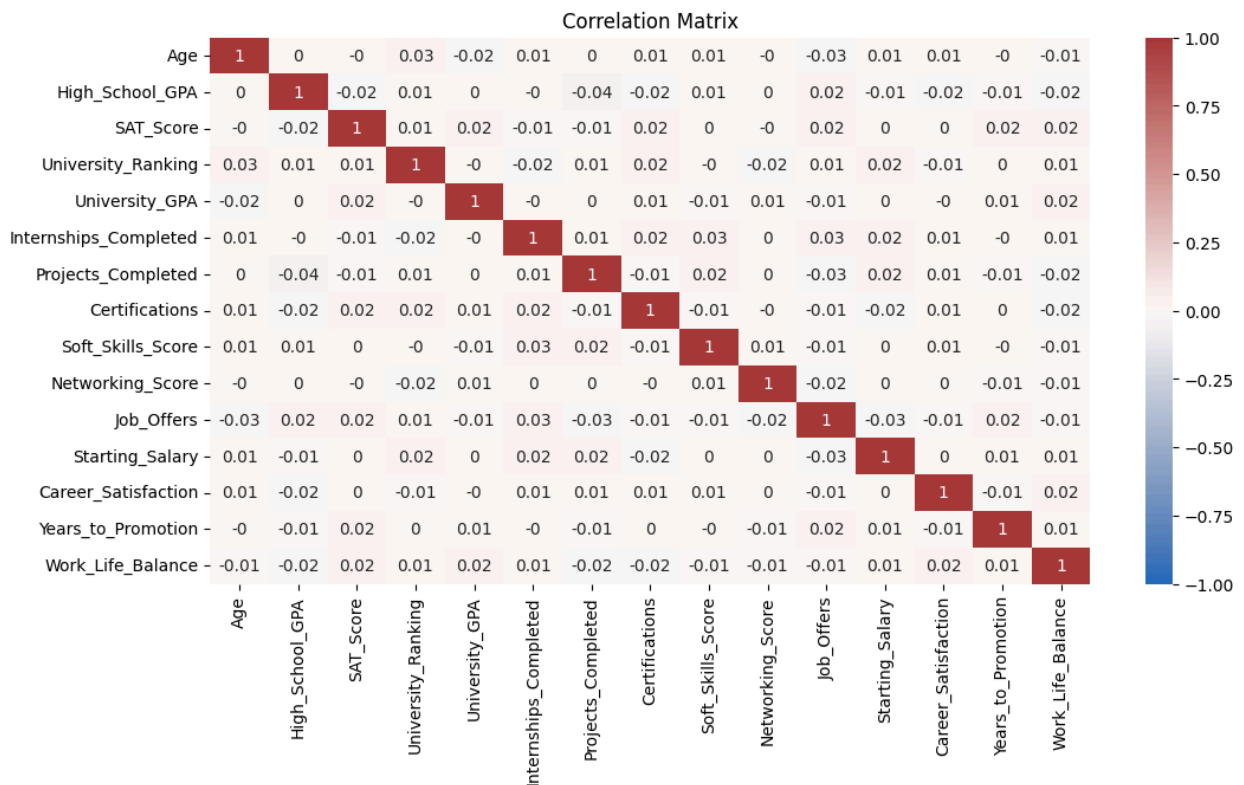
```
Out[6]: Text(0.5, 1.0, 'Internships Completed vs. Job Offers')
```



The scatter plot shows if there was a relationship between *Internships Completed* and *Job Offers*. As seen on the scatter plot, it is very difficult to see if there is a positive or negative relationship since having completed 1 - 4 internships doesn't show if Job Offers increased or decreased. For further analysis a correlation matrix is our next step to see how different variables are correlated to each other and how strong their correlation is.

```
In [7]: # Correlation Matrix
corr_matrix = df.corr(numeric_only=True).round(2)
plt.figure(figsize=(12,6))
sns.heatmap(corr_matrix, annot=True, vmax=1, vmin=-1, cmap='vlag')
plt.title('Correlation Matrix')
```

```
Out[7]: Text(0.5, 1.0, 'Correlation Matrix')
```



Looking at the correlation matrix performed on the data set we can see how for the most part a majority of the graph is marked by low values and when looking closely at the job offers section and lining it up with internships completed the value is still low at .03 implying that number of internships completed has a really weak, positive correlation to job offers received. Next, we would have to run predictive analysis to see whether *Internships Completed* implies causation to *Jobs Offered*.

Predictive Analytics

Predictive Analytics, in specific a simple linear regression or Ordinary Least Squared (OLS) regression will be used to see if our independent variable, *Internships Completed* is statistically significant to the independent variable *Job Offers*.

We, first define our dependent and independent variables, X and Y. Then, we create our OLS model using *statsmodels.api*

```
In [11]: X = df['Internships_Completed']
Y = df['Job_Offers']

X = sm.add_constant(X) # Adds a constant or intercept to the model

model = sm.OLS(Y,X).fit()
print(model.summary())
```

OLS Regression Results						
Dep. Variable:	Job_Offers	R-squared:	0.001			
Model:	OLS	Adj. R-squared:	0.001			
Method:	Least Squares	F-statistic:	5.014			
Date:	Wed, 14 May 2025	Prob (F-statistic):	0.0252			
Time:	18:57:51	Log-Likelihood:	-9779.6			
No. Observations:	5000	AIC:	1.956e+04			
Df Residuals:	4998	BIC:	1.958e+04			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	2.4125	0.042	57.734	0.000	2.331	2.494
Internships_Completed	0.0385	0.017	2.239	0.025	0.005	0.072
Omnibus:	8520.338	Durbin-Watson:	1.989			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	332.312			
Skew:	0.013	Prob(JB):	6.91e-73			
Kurtosis:	1.737	Cond. No.	4.70			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

OLS Regression Results

There are several factors that we are going to interpret from the OLS Regression Results.

- The First is *Prob (F-statistic)*, which is 0.0252. Since it is less than 0.05 significance level, then the whole model is significant.
- Next is *R-squared*, which says that 0.1% of the dependent variable, Job Offers, can be explained by the independent variable, Internships Completed.
- the *P>|t|*, which is the P-value lets us know if the independent variable has a statistically significant relationship to the dependent variable. Since 0.025 is less than the significance level of 0.05, then there is a statistically significant relationship and therefore we can reject the null hypothesis H_0 .
- The values from the *coef* column are used to create the regression equation. It also states that having 0 internships completed, on average, each student gets 2.4125 Job offers (const) value. Also, that holding everything else constant, for each internship completed, your chances of obtaining a Job Offer increases by 3.85%.

$$\text{Job_Offers_hat} = 2.4125 + 0.0385(\text{Internships_Completed})$$

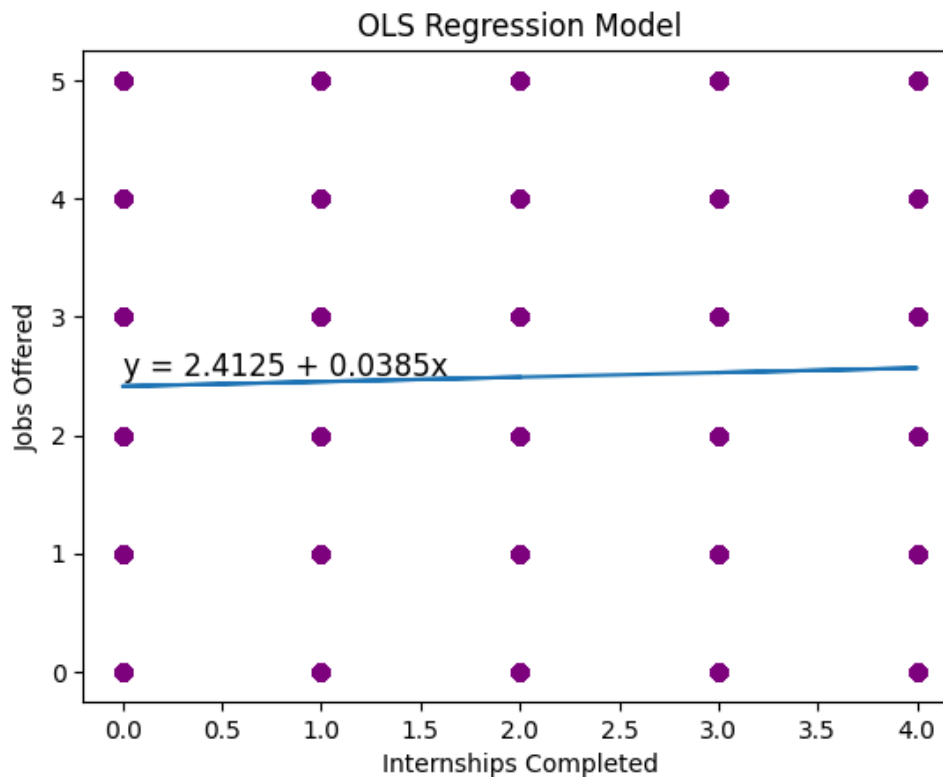
```
In [9]: # Visualize the Best Fit
import numpy as np

a, b = np.polyfit(df['Internships_Completed'], df['Job_Offers'], 1) # Find line of best fit
plt.scatter(df['Internships_Completed'], df['Job_Offers'], color='purple') # Adds points to plot
plt.plot(df['Internships_Completed'], a*df['Internships_Completed'] + b) # Add line of best fit

plt.text(0, 2.5, 'y = ' + '{:.4f}'.format(b) + ' + {:.4f}'.format(a) + 'x', size = 12) # Add equation to plot

# Add axis labels and title
plt.title('OLS Regression Model')
plt.xlabel('Internships Completed')
plt.ylabel('Jobs Offered')
```

```
Out[9]: Text(0, 0.5, 'Jobs Offered')
```



Summary

In conclusion, the purpose of our project was to analyze the Education Career Success dataset and see what contributing factors can attribute towards the success of someone receiving a job offer paying keen attention to the internships completed. Our null hypothesis, infers that one or more completed internships does not have an impact on obtaining a job offer while the alternative hypothesis states that one or more completed internships does have an impact on obtaining a job offer. When conducting our descriptive analytics we noticed through the use of the describe function alongside a box plot graph how factors such as field of study don't play a factor into the amount of job offers someone is to receive. We then conducted analysis of variance to solidify our results proving that field of study plays no statistically significant role in job offers received. We then performed a correlation matrix to see what factors are to play the biggest role in job offers obtained with the stand out being internships completed. With our predictive analytics we wanted to test how statistically significant internships completed was which we accomplished through the use of OLS regression. Analyzing the results of our OLS regression we came to the conclusion that internships completed has a statistically significant relationship with job offers thus allowing us to reject our null hypothesis.

- **Dataset** [education_career_success.csv](#) was obtained from [Kaggle](#)
- **Markdown** syntax and writing from [GitHub Docs](#)
- **OLS Regression** code reference from [Statology](#)